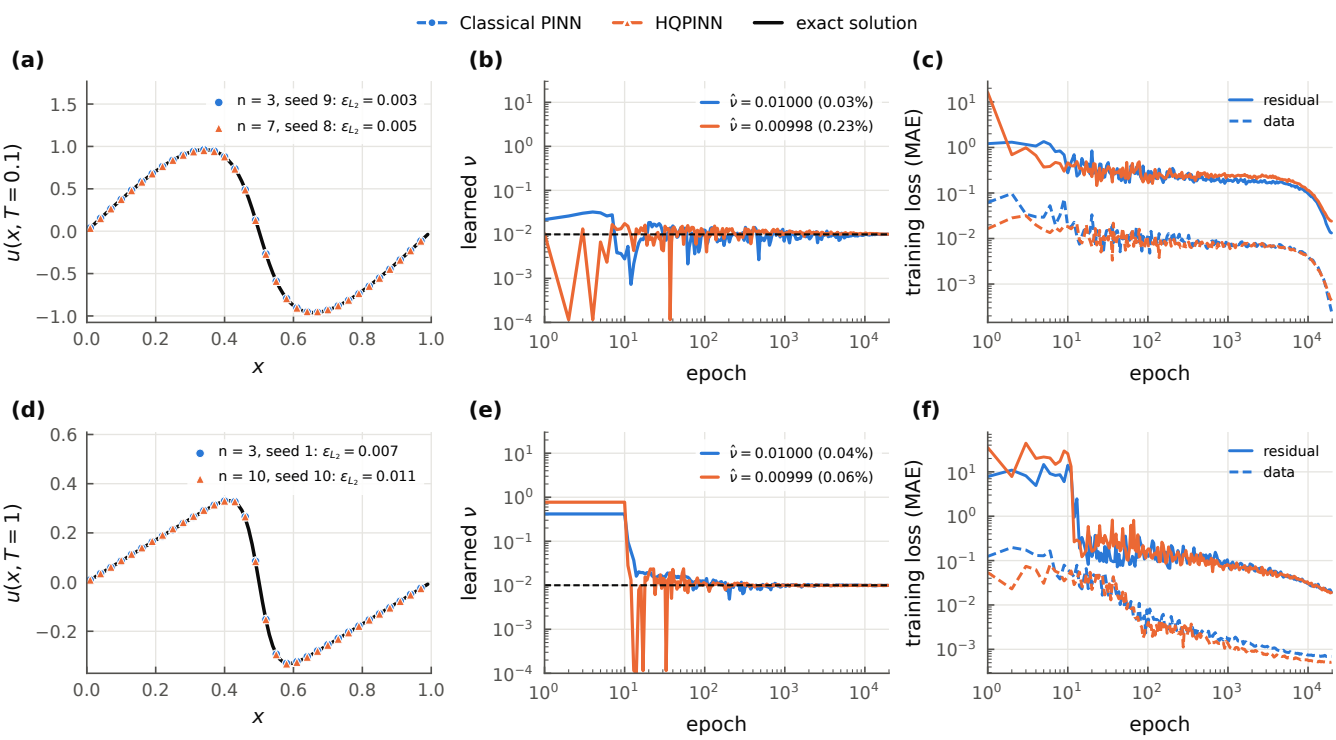

Classical PINN
HQPINN
exact solution
(a)
n = 3, seed 9: $\varepsilon_{L_2} = 0.003$
n = 7, seed 8: $\varepsilon_{L_2} = 0.005$
$u(x, T = 0.1)$
x
(b)
$\hat{\nu} = 0.01000$ (0.03%)
$\hat{\nu} = 0.00998$ (0.23%)
learned ν
epoch
(c)
residual
data
training loss (MAE)
epoch
(d)
n = 3, seed 1: $\varepsilon_{L_2} = 0.007$
n = 10, seed 10: $\varepsilon_{L_2} = 0.011$
$u(x, T = 1)$
x
(e)
$\hat{\nu} = 0.01000$ (0.04%)
$\hat{\nu} = 0.00999$ (0.06%)
learned ν
epoch
(f)
residual
data
training loss (MAE)
epoch